

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850 . In triangle XYZ , the length of $\overline{YZ}$ is 60 . What is the length of $\overline{XY}$?

Answer

- A. 204
- B. 182
- C. 60
- D. 12

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. It follows that side $\overline{AB}$ corresponds to side $\overline{XY}$ and side $\overline{BC}$ corresponds to side $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{XY}{AB} = \frac{YZ}{BC}$. Substituting 170 for AB , 60 for YZ , and 850 for BC in this equation yields $\frac{XY}{170} = \frac{60}{850}$. Multiplying each side of this equation by 170 yields $XY = 12$. Therefore, the length of $\overline{XY}$ is 12 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of $\overline{YZ}$, not $\overline{XY}$.